

TECHNICAL ENGINEERING & IMPLEMENTATION REPORT

WINDOWS SERVER ARCHITECTURE

& Enterprise IT Infrastructure

An in-depth deployment of MCSA Directory Services,
Access Governance, and Core Routing Integration (CCNA)

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1 Project Overview, Architectural Scope & Enterprise Topology

1.1 Executive Summary & Context

Deploying Microsoft Directory Services in enterprise production environments requires absolute convergence with the underlying network fabric. Routing asymmetries, misconfigured broadcast forwarders, or improper name resolution boundaries directly compromise directory replication, authentication latency, and failover resilience.

This engineering report documents the end-to-end design, implementation, and stress-testing of a fault-tolerant corporate infrastructure virtualized inside **PNETLab**. The architecture bridges core Cisco routing and switching (CCNA principles) with enterprise Windows Server system administration (MCSA Directory Services). The deployment spans multi-VLAN segmentation, cross-subnet dynamic addressing, automated Group Policy governance, multi-site replication, high-availability DHCP clustering, and a fully remediated Disaster Recovery scenario driven by unrecoverable primary controller hardware failure.

1.2 Enterprise PNETLab Network Topology

The complete virtualized enterprise environment was constructed inside PNETLab, integrating Cisco virtual IOS appliances with Windows Server instances and Windows 11 desktop endpoints. Figure 1 details the physical and logical interconnectivity across all tiers.

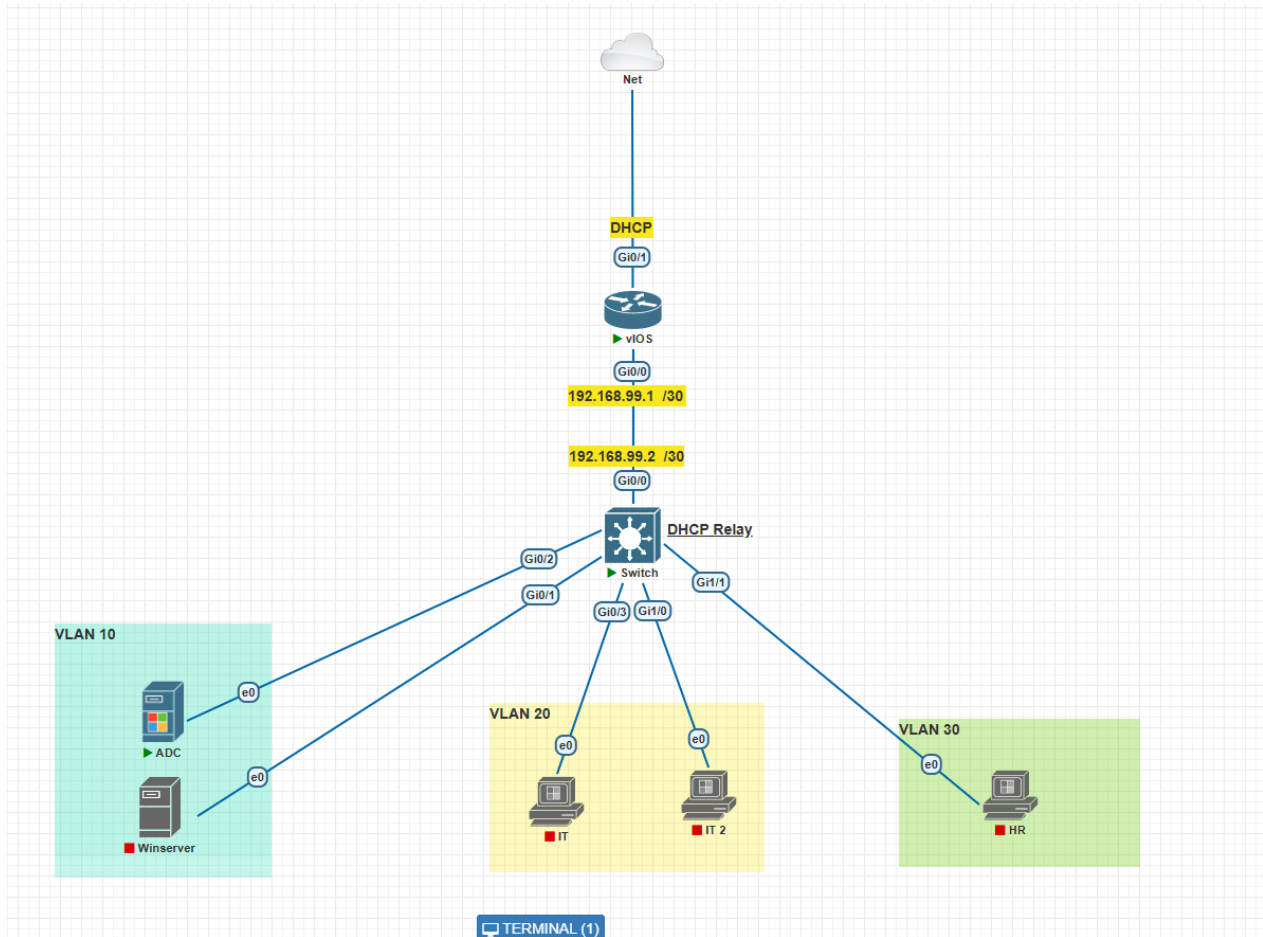


Figure 1: End-to-End Enterprise Lab Topology in PNETLab: Cisco Edge Router, Layer 3 Core Switch, Multi-VLAN Segmentation, Redundant Domain Controllers, and Segmented Client Workstations.

1.3 Core Engineering Pillars & Scope of Implementation

The implementation encompasses five distinct technical disciplines executed in chronological phases:

1. Layer 3 Switching, Transit Routing & Network Address Translation:

- Segmentation of broadcast domains into dedicated VLANs: Servers (VLAN 10), IT Clients (VLAN 20), and Human Resources (VLAN 30).
 - Inter-VLAN routing terminated on Layer 3 Switch Virtual Interfaces (SVIs) with Spanning Tree portfast edge-port acceleration.
 - Implementation of Cisco `ip helper-address` forwarders on client SVIs to unicast cross-subnet DHCP broadcast traffic (UDP 67) to central servers while injecting gateway identity.
 - Routed point-to-point transit link (/30) to a Cisco Edge Router with symmetric static return routing and NAT Overload (PAT) boundary provisioning.
2. **Directory Baseline, Name Resolution & Centralized Identity Governance:**
- Installation of Active Directory Domain Services (AD DS) provisioning the forest root domain (`amazon.local`).
 - Authoritative Active Directory-Integrated DNS architecture featuring forward lookup zones, domain locator SRV records (`_msdcs`), reverse lookup zones (`10.168.192.in-addr.arpa`), service aliases (CNAME), and external upstream forwarders.
 - Multi-tier Organizational Unit (OU) structure isolating departmental identities, coupled with Global Security Groups (IT-Staff, HR-Staff) to enforce access boundaries.
 - Centralized departmental file storage (`C:\Shares`) enforcing explicit zerotrust NTFS permissions by severing inheritance, purging default BUILTIN\Users DACLs, and bypassing SMB alias restrictions via LanmanServer registry tuning (`DisableStrictNameChecking`).
3. **Desktop Standardization & Automated Software Deployment via GPO:**
- Enforcement of administrative security baselines: disabling the Run dialogue (`Win + R`) for IT personnel and restricting Control Panel/Windows Settings access for HR personnel.
 - Central software repository provisioning with authenticated network share permissions.
 - Machine-targeted silent deployment of 7-Zip (.MSI) during pre-logon system startup executing under the privileged NT AUTHORITY\SYSTEM context.
4. **Hypervisor Clone Remediation, Secondary Controller Provisioning & High Availability:**
- Resolution of virtual machine cloning conflicts (Machine SID duplication) by manipulating the Windows Sysprep state-machine via registry modifications (`GeneralizationState`, `CleanupState`, `SkipRearm`) to bypass fatal generalization crashes.
 - Promotion of an Additional Domain Controller (ADC) with Global Catalog (GC) and DNS replication, synchronized within the 5-minute Kerberos time-skew tolerance.
 - Geographic site partitioning in Active Directory Sites and Services into HQ (`Amman`) and Branch (`Irbid`) linked via `DEFAULTIPSITELINK`.
 - Activation of the Active Directory Recycle Bin at the 2016 functional baseline, validating attribute retention across the 180-day `msDS-deletedObjectLifetime` window without requiring DSRM offline restores.
 - Deployment of active/passive DHCP Failover in Hot Standby mode across TCP 647, featuring a 5% Reserve Pool, a 10-minute State Switchover Interval, and redundant `ip helper-address` configurations on core switch SVIs.
5. **Catastrophic Disaster Recovery, FSMO Seizure & Core Services Remediation:**
- Simulation of sudden, unrecoverable hardware destruction by hard power-cycling the Primary Domain Controller (PDC1).
 - Bypassing graphical RPC timeout stalls by executing forced PowerShell role seizure across all five Flexible Single Master Operation (FSMO) roles (`Schema`, `Domain Naming`, `PDC Emulator`, `RID`, `Infrastructure Master`) on ADC.
 - End-to-end directory metadata cleanup via Active Directory Users and Computers (ADUC), Active Directory Sites and Services, and DNS Manager (purging stale A, NS, and CNAME records).
 - Core services realignment: updating DHCP Scope Option 006 (DNS Servers) to redistribute the surviving controller, validating dynamic IP acquisition under the 1-hour Maximum Client Lead Time lease condition, and verifying directory write readiness via RID pool issuance.

1.4 Infrastructure Systems & Addressing Matrix

Table 1 details the complete hardware-virtualization abstraction layer, operating systems, and service assignments implemented throughout the topology.

Node Identifier	Platform / OS	Subnet / VLAN	Assigned IP	Hosted Services & Enterprise Roles
R1-Edge	Cisco vIOS-L3	Transit / WAN	192.168.99.1	Edge Gateway, PAT Overload, Return Routing
SW-Core	Cisco vIOS-L2/L3	Infrastructure	192.168.99.2	Inter-VLAN Routing (SVIs), Dual DHCP Relays
DC01	Windows Server 2022	VLAN 10 (Servers)	192.168.10.10	Primary DC, FSMO Holder, Authoritative DNS, Primary DHCP
ADC	Windows Server 2022	VLAN 10 (Servers)	192.168.10.11	Additional DC, Global Catalog, Standby DHCP, DR Target
PC-IT1	Windows 11 Pro	VLAN 20 (Clients)	Dynamic (DHCP)	Workstation Endpoint, GPO Validation Node
PC-IT2	Windows 11 Pro	VLAN 20 (Clients)	Dynamic (DHCP)	Failover Verification Client, MCLT Validation
PC-HR	Windows 11 Pro	VLAN 30 (HR)	Dynamic (DHCP)	Isolated Departmental Client, Access Audit Node

Table 1: Comprehensive breakdown of infrastructure hosts, addressing architecture, and deployed system roles.

2 Network Layout and Subnetting

This phase establishes the underlying routing, VLAN isolation, and Internet access required before deploying Windows Server domain services. The core Layer 3 switch handles local inter-VLAN routing, and the edge router provides NAT overloading (PAT) to the simulated Internet cloud.

The server infrastructure segment (VLAN 10) is pre-provisioned to accommodate both the primary Domain Controller (DC01) and a dedicated Additional Domain Controller (DC02-ADC) within the same broadcast domain.

Subnet	Network ID	Default Gateway	Node Role / IP Range	VLAN
Servers	192.168.10.0/24	192.168.10.1 (SVI)	DC01 (.10), DC02-ADC (.11)	VLAN 10
Clients (IT)	192.168.20.0/24	192.168.20.1 (SVI)	DHCP Range (.50 – .150)	VLAN 20
Human Resources	192.168.30.0/24	192.168.30.1 (SVI)	DHCP Range (.50 – .100)	VLAN 30
Transit Link	192.168.99.0/30	—	.1 (Router) / .2 (L3 Switch)	Routed
WAN / Cloud	Dynamic (ISP)	ISP Gateway	Router Gi0/0 (DHCP / PAT)	Outside

Table 2: Allocated IP subnets and gateway assignments.

3 Applied Device Configurations

3.1 Layer 3 Core Switch

VLANs were created and terminated on switch virtual interfaces (SVIs) to handle inter-VLAN routing locally. Physical access ports were assigned to their respective VLANs, including dedicated server ports on VLAN 10 for DC01 (ga0/3) and DC02 (ga0/5). `ip helper-address` was applied to VLAN 20 and VLAN 30 to forward client DHCP requests across subnets to the primary server.

```
1 ! VLAN and Routing Activation
2 ip routing
3 vlan 10
4   name SERVERS
5 vlan 20
6   name CLIENTS
7 vlan 30
8   name HR
9 !
10 ! Gateway SVIs & DHCP Relay
11 interface Vlan10
12   ip address 192.168.10.1 255.255.255.0
13   no shutdown
14 !
15 interface Vlan20
16   ip address 192.168.20.1 255.255.255.0
17   ip helper-address 192.168.10.10
18   no shutdown
19 !
20 interface Vlan30
21   ip address 192.168.30.1 255.255.255.0
22   ip helper-address 192.168.10.10
23   no shutdown
24 !
25 ! Server Access Ports (Primary DC & ADC)
26 interface g0/1
27   description Windows-Server-DC01
28   switchport mode access
29   switchport access vlan 10
30   spanning-tree portfast
31 !
32 interface g0/2
33   description Windows-Server-DC02-ADC
34   switchport mode access
```

```

35 switchport access vlan 10
36 spanning-tree portfast
37 !
38 ! Client & HR Access Interfaces
39 interface range g0/3
40 switchport mode access
41 switchport access vlan 20
42 spanning-tree portfast
43 !
44 interface range g1/0
45 switchport mode access
46 switchport access vlan 20
47 spanning-tree portfast
48 !
49 interface g1/1
50 switchport mode access
51 switchport access vlan 30
52 spanning-tree portfast
53 !
54 ! Routed Transit Interface & Default Route
55 interface g0/0
56 no switchport
57 ip address 192.168.99.2 255.255.255.252
58 no shutdown
59 !
60 ip route 0.0.0.0 0.0.0.0 192.168.99.1

```

3.2 Edge Router (Static Routing & NAT)

Static routes were added on the router for internal subnets pointing to the L3 switch, and PAT was configured on the outside interface.

```

1 ! Interfaces
2 interface Gi0/1
3 ip address dhcp
4 ip nat outside
5 no shutdown
6 !
7 interface Gi0/0
8 ip address 192.168.99.1 255.255.255.252
9 ip nat inside
10 no shutdown
11 !
12 ! Return Routes to Internal Subnets
13 ip route 192.168.10.0 255.255.255.0 192.168.99.2
14 ip route 192.168.20.0 255.255.255.0 192.168.99.2
15 ip route 192.168.30.0 255.255.255.0 192.168.99.2
16 !
17 ! NAT Overload (PAT)
18 access-list 1 permit 192.168.10.0 0.0.0.255
19 access-list 1 permit 192.168.20.0 0.0.0.255
20 access-list 1 permit 192.168.30.0 0.0.0.255
21 ip nat inside source list 1 interface Gi0/0 overload

```

4 Verification & Screenshots

4.1 Layer 3 Switch Interfaces

The command `show ip interface brief` confirms that all SVIs and the transit interface are active and operational.

```

Switch(config-if)#
Switch(config-if)#show ip int
Switch(config-if)#end
Switch#sho
*Sep  9 18:08:53.968: %SYS-5-CONFIG_I: Configured from console by consolew
Switch#show ip int
Switch#show ip interface br
Switch#show ip interface brief

```

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet0/1	unassigned	YES	unset	up	up
GigabitEthernet0/2	unassigned	YES	unset	up	up
GigabitEthernet0/3	unassigned	YES	unset	up	up
GigabitEthernet0/0	192.168.99.2	YES	manual	up	up
GigabitEthernet1/0	unassigned	YES	unset	up	up
GigabitEthernet1/1	unassigned	YES	unset	up	up
GigabitEthernet1/2	unassigned	YES	unset	down	down
GigabitEthernet1/3	unassigned	YES	unset	down	down
Vlan10	192.168.10.1	YES	manual	up	up
Vlan20	192.168.20.1	YES	manual	up	up
Vlan30	192.168.30.1	YES	manual	up	up

```

Switch#

```

Figure 2: Layer 3 Switch interface status.

4.2 Router Routing Table and NAT Output

The routing table confirms return paths to internal networks, and the NAT table confirms active translation sessions when hosts access external resources.

```

Gateway of last resort is 10.0.137.1 to network 0.0.0.0

S*   0.0.0.0/0 [254/0] via 10.0.137.1
      10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C     10.0.137.0/24 is directly connected, GigabitEthernet0/1
L     10.0.137.1/32 is directly connected, GigabitEthernet0/1
S     192.168.10.0/24 [1/0] via 192.168.99.2
S     192.168.20.0/24 [1/0] via 192.168.99.2
S     192.168.30.0/24 [1/0] via 192.168.99.2
      192.168.99.0/24 is variably subnetted, 2 subnets, 2 masks
C     192.168.99.0/30 is directly connected, GigabitEthernet0/0
L     192.168.99.1/32 is directly connected, GigabitEthernet0/0

```

Figure 3: Edge router routing table.

5 Implementation Notes

- **Symmetric Routing for NAT:** The router must have explicit static routes back to the internal subnets via the L3 switch (192.168.99.2). Without these routes, inbound reply packets from the Internet are dropped before the router can de-translate them.
- **DHCP Helper Placement:** The `ip helper-address` command is configured on the client SVIs (Vlan20 and Vlan30) rather than on the server SVI. It converts client broadcast discovery packets into unicast traffic directed at 192.168.10.10 and populates the GIADDR field so the server leases from the correct scope.
- **Server Infrastructure Co-location:** Hosting both DC01 and DC02-ADC within the same VLAN allows replication traffic to remain entirely at Layer 2, avoiding router/SVI processing overhead during directory synchronization.

6 Windows Server Provisioning and AD DS Promotion

The server host was integrated into VLAN 10 with static addressing before role deployment. Active Directory Domain Services (AD DS) was installed to provision the forest root domain `it.local`.

Configuration Parameter	Value Assigned	Operational Role / Notes
Hostname	DC01	Primary Domain Controller identifier
IP Address / Netmask	192.168.10.10 /24	Static identity within Server Farm segment (VLAN 10)
Default Gateway	192.168.10.1	L3 Switch SVI interface for inter-VLAN transit
Preferred DNS Resolver	127.0.0.1	Loopback address querying local directory SRV records
Active Directory Domain	amazon.local	Kerberos realm and administrative security boundary
Forest Functional Level	Windows Server 2016	Domain interoperability baseline
DSRM Recovery Password	Configured	Local credential for offline NTDS database maintenance

Table 3: Host network parameters and directory services promotion baseline.

7 Core Infrastructure Services: DNS and DHCP Architecture

7.1 DNS Architecture and Resolution Mechanisms

The integrated DNS Server role handles internal directory service resolution and external Internet lookup:

- **Forward Lookup Zone:** Authoritative for `amazon.local`, holding dynamic A records and domain locator SRV records under `_msdcs.amazon.local`.
- **Reverse Lookup Zone:** An Active Directory-integrated zone (`10.168.192.in-addr.arpa`) was initialized with secure dynamic updates for PTR record maintenance.
- **Service Alias (CNAME):** An alias record `files.amazon.local` was mapped to `DC01.amazon.local` to decouple file share access paths from physical hostnames.
- **External Forwarders:** Configured with public DNS server `8.8.8.8` to resolve non-authoritative queries via the edge router NAT boundary.

7.2 Multi-Subnet DHCP Architecture & Cross-VLAN Relay

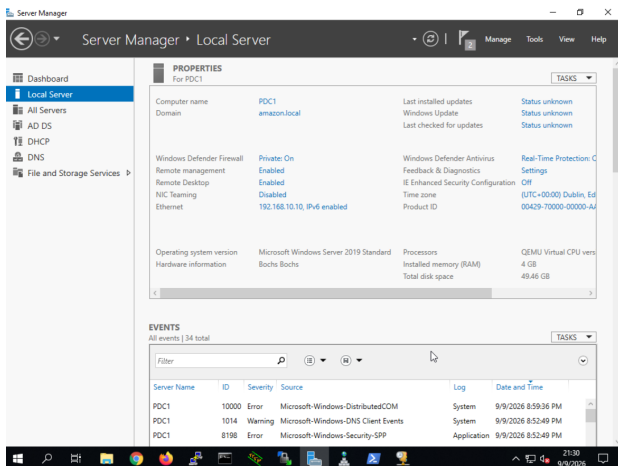
The DHCP role was deployed, authorized in Active Directory, and coupled with the L3 Switch relay infrastructure:

Parameter	Scope: CLIENTS-VLAN20	Scope: HR-VLAN30	Scope Option ID
Subnet Scope ID	192.168.20.0 /24	192.168.30.0 /24	—
Distribution Pool	192.168.20.50 – .150	192.168.30.50 – .100	—
Router (Gateway)	192.168.20.1 (SVI)	192.168.30.1 (SVI)	Option 003
DNS Server	192.168.10.10	192.168.10.10	Option 006
DNS Domain Name	amazon.local	amazon.local	Option 015
Lease Duration	8 Days	8 Days	Default

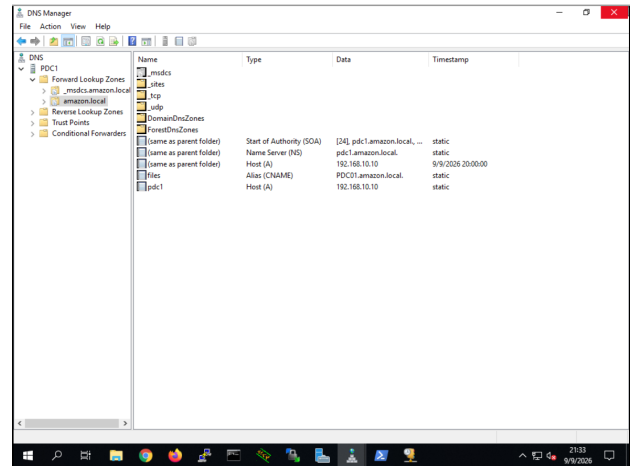
Table 4: DHCP multi-subnet pool allocations and network options.

8 Core Services Verification

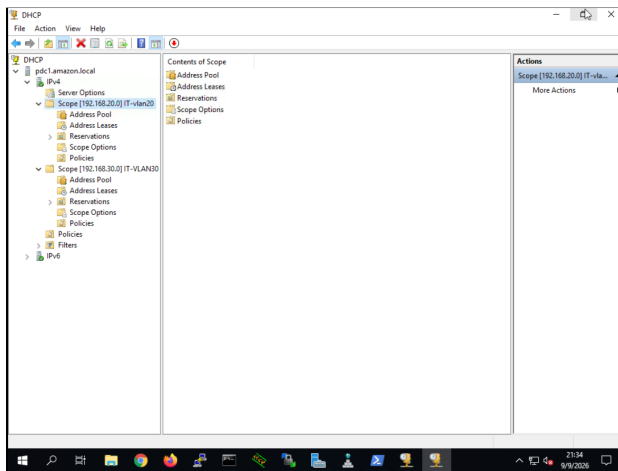
The 2×2 grid below documents the operational verification of host identity, DNS zone configuration, authorized DHCP scopes, and cross-subnet dynamic lease allocation.



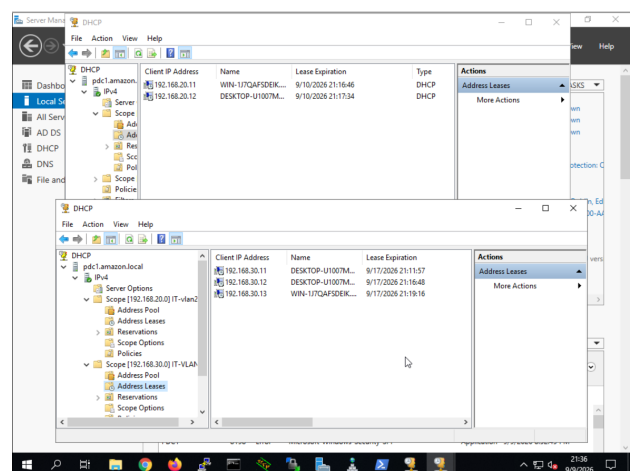
(a) Host identity and domain baseline.



(b) Forward, reverse, and alias DNS records.



(c) Authorized and operational DHCP scopes.



(d) Dynamic address allocation via relay.

Figure 4: Windows Server core services verification

9 Directory Architecture, Identity Scoping, and Workstation Onboarding

Directory assets were organized into dedicated departmental Organizational Units (OUs) beneath an enterprise root container to isolate policy boundaries and streamline administrative delegation. Access governance is enforced exclusively through Global Security Groups.

Parent OU	Department OU	Security Group	Assigned Domain Assets
Company	IT	IT-Staff	it_user1, it_user2, PC1, PC2
Company	HR	HR-Staff	hr_user1, PC-HR

Table 5: Organizational hierarchy, security groups, and provisioned domain assets.

- **Workstation Domain Integration:** Workstations across VLAN 20 and VLAN 30 were provisioned with unique NetBIOS identifiers and joined to `amazon.local` via secure Netlogon channels. Workstations locate domain controllers using DNS SRV records (`_ldap._tcp.dc._msdcs`) resolved by the domain controller.
- **Container Optimization:** Computer accounts were relocated from the default `CN=Computers` container into their target departmental OUs, ensuring direct inheritance of machine-scoped Group Policy Objects.

10 Departmental Storage and Discretionary Access Control

Centralized storage was established on the primary domain controller under `C:\Shares`. Network boundary protection is managed through SMB share permissions, while granular authorization is enforced by the NTFS Discretionary Access Control List (DACL).

Target Folder	Share Permission	NTFS Permission Entry	Effective Authorization
\Shares	Authenticated Users: Change	Administrators: Full Control	Perimeter transit gateway
\IT_Data	Inherited from parent	IT-Staff: Modify	Authorized IT modify/read/write
\HR_Data	Inherited from parent	HR-Staff: Modify	Authorized HR modify/read/write

Table 6: Share permissions versus NTFS DACL departmental isolation matrix.

- **Inheritance Disruption:** Inheritance was severed on departmental folders, converting parent permissions into explicit entries. Generic `BUILTIN\Users` permissions were purged, establishing strict zero-trust isolation.
- **SMB Alias Integration:** The server was configured to accept alias-targeted UNC traffic by setting `DisableStrictNameChecking` to 1 and registering the alias `files` under `OptionalNames` in the `LanmanServer` parameters. Workstations access the storage via `\\files.amazon.local\Shares`.

11 Group Policy Enforcement and Automated Software Deployment

Administrative Templates and Software Installation extensions were configured to apply security baselines and standardize client software environments without manual workstation configuration.

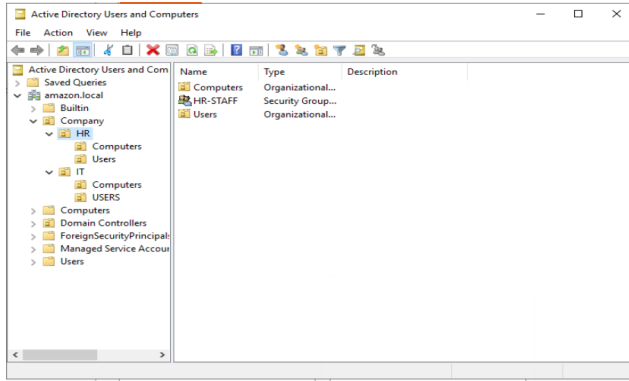
GPO Identifier	Scope	Target Container	Enforced Baseline Policy
GPO-Disable-Run	User Configuration	IT OU	Prohibits Run dialog and shortcut keys
GPO-Block-ControlPanel	User Configuration	HR OU	Restricts Control Panel and Settings access
GPO-Deploy-7Zip	Computer Configuration	Department Workstations	Silent MSI package installation at boot

Table 7: Group Policy implementation scope, target containers, and policy goals.

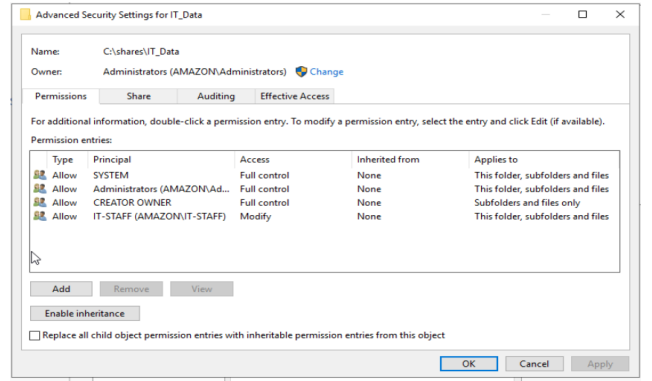
- **Automated Packaging Architecture:** A central distribution share was provisioned at `\\DC01.amazon.local\Software` with read access granted to Domain Computers.
- **Assigned Deployment Model:** The 7-Zip package was assigned under Computer Configuration using a strict UNC network path. Workstations process the Windows Installer package during pre-logon boot under the NT AUTHORITY\SYSTEM context.

12 Implementation Verification Evidence

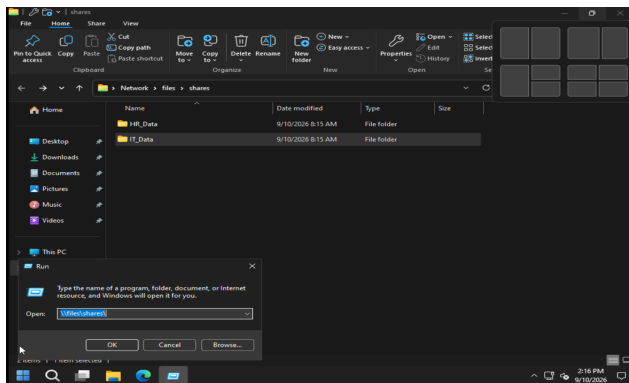
The operational verification of directory hierarchy, access control enforcement, network storage resolution, and policy deployment is summarized below.



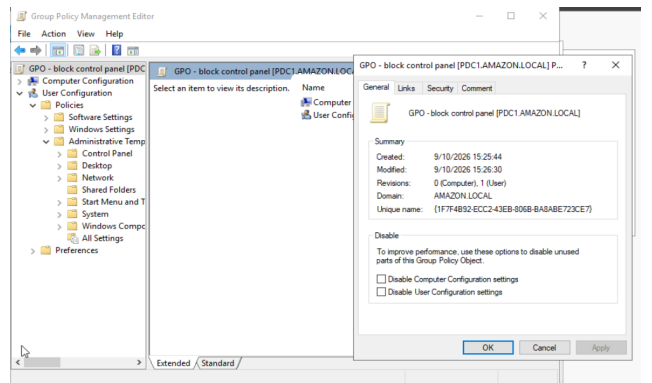
(a) Directory and OU structure.



(b) NTFS permissions and DACL.



(c) SMB access and isolation.



(d) GPO deployment configuration.

Figure 5: Active Directory, storage access control, and policy verification

13 Additional Domain Controller (ADC) Deployment

13.1 Introduction and Objectives

In enterprise Active Directory environments, deploying an Additional Domain Controller (ADC) is essential for eliminating single points of failure. The primary objective of configuring an ADC within the `amazon.local` domain is to achieve high availability, distribute authentication workloads (Kerberos/NTLM), ensure local DNS resilience, and maintain a synchronized replica of the directory database (`NTDS.dit`). Furthermore, the ADC serves as a hot-standby authority capable of assuming operations master roles during disaster recovery scenarios.

13.2 Prerequisite: Resolving Machine SID Duplication

Before joining the server to the domain and promoting it to an ADC, an architectural conflict resulting from virtual machine cloning must be resolved: **Machine Security Identifier (SID) duplication**.

13.2.1 Why SID Regeneration is Mandatory

Windows uses a unique Machine SID as the cryptographic base for all local and domain security principals. When an existing virtual machine is cloned in a virtualized lab (e.g., PNETLab), the clone inherits the identical Machine SID of the parent machine. Attempting to join this clone to `amazon.local` triggers an explicit security validation failure:

"The domain join cannot be completed because the SID of the domain you attempted to join was identical to the SID of this machine."

To resolve this, the system must undergo generalization via the System Preparation tool (`sysprep.exe`) to strip local identifiers and generate a unique Machine SID.

13.2.2 Bypassing Sysprep Fatal Error via Registry

Executing `sysprep.exe` with the `/generalize` flag often terminates with: *"A fatal error occurred while trying to sysprep the machine."* This issue stems from incomplete cleanup states or exhaustion of the Windows licensing re-arm counter. To override the state machine, three registry parameters must be configured manually in `regedit`:

- **Generalization State:** `HKLM\SYSTEM\Setup\Status\SysprepStatus`
Set `GeneralizationState` to 7 (specifies that the system is ready for generalization).
- **Cleanup State:** `HKLM\SYSTEM\Setup\Status\SysprepStatus`
Set `CleanupState` to 2 (indicates that previous cleanup routines succeeded).
- **Skip Rearm:** `HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion\SoftwareProtectionPlatform`
Set `SkipRearm` to 1 (bypasses software license re-arm limitations).

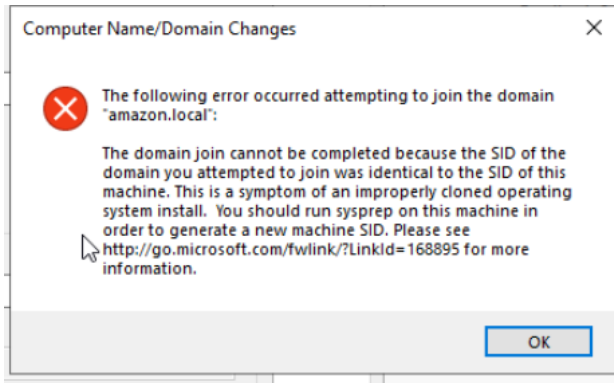
13.2.3 Execution and SID Verification

With the registry states corrected, the generalization routine is executed from an administrative command prompt:

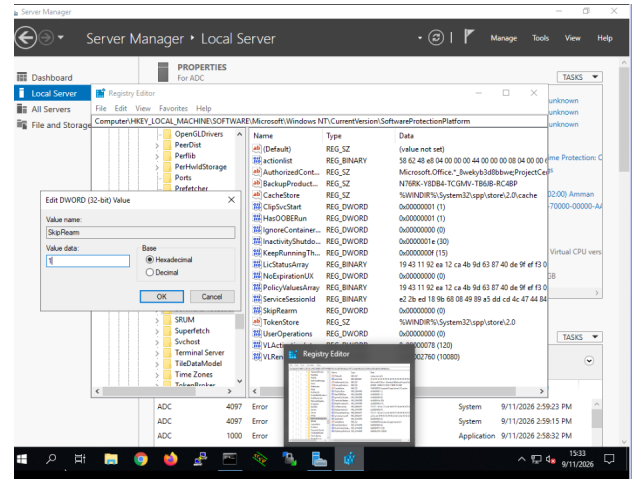
```
C:\Windows\System32\Sysprep\sysprep.exe /oobe /generalize /reboot
```

The `/generalize` parameter removes hardware abstraction layer bindings and local SIDs, while `/oobe` triggers the Out-of-Box Experience on reboot to regenerate a unique Machine SID. Following the restart, the newly generated SID is verified using:

```
whoami /user
```



(a) Sysprep fatal error during generalization.



(b) Registry keys override in SysprepStatus.

Figure 6: Diagnosing and resolving the Sysprep fatal error via registry modifications.

14 Host Preparation, Domain Joining, and Promotion

14.1 Network Configuration and Host Identification

Following the regeneration of the Machine SID, the secondary server was assigned a static IPv4 configuration (192.168.10.11/24) with the default gateway set to 192.168.10.1. The Preferred DNS server was directed to the Primary Domain Controller (192.168.10.10) to allow directory locator service resolution. The host NetBIOS name was updated to ADC. In addition, the system clock and time zone were strictly aligned with the Primary DC to meet the maximum 5-minute skew tolerance required by the Kerberos v5 authentication protocol.

14.2 Domain Integration and AD DS Promotion

The host was formally joined to the `amazon.local` domain using domain administrative credentials and rebooted. Following authentication into the domain, the **Active Directory Domain Services (AD DS)** role was installed via Server Manager, and the post-deployment promotion wizard was configured with the following parameters:

- **Deployment Configuration:** Selected *"Add a domain controller to an existing domain"* targeting `amazon.local`.
- **Domain Controller Capabilities:** Checked both **Domain Name System (DNS) server** and **Global Catalog (GC)** to ensure local query resolution and autonomous logon authentication.
- **DSRM Credentials:** Specified a Directory Services Restore Mode administrative recovery password.
- **Replication Partner:** Designated the Primary DC (`DC01.amazon.local`) as the replication source for initial partition synchronization.

After passing all prerequisite checks, the promotion was executed. The server completed its initialization of `NTDS.dit` and `SYSVOL`, rebooted, and joined the topology as a fully functional, writable Additional Domain Controller.

15 Active Directory Sites and Services Configuration

To optimize replication schedules and localize authentication traffic across physical boundaries, the directory topology was structured into two geographic sites using **Active Directory Sites and Services**. The default site (`Default-First-Site-Name`) was renamed to `Amman`, designating the primary headquarters where DC01 resides. A secondary site was created under the name `Irbid` and bound to the `DEFAULTIPSITELINK` inter-site transport. Finally, the ADC object was relocated from `Sites\Amman\Servers` into the `Irbid` site container to represent the remote branch controller, as illustrated in Figure 7.

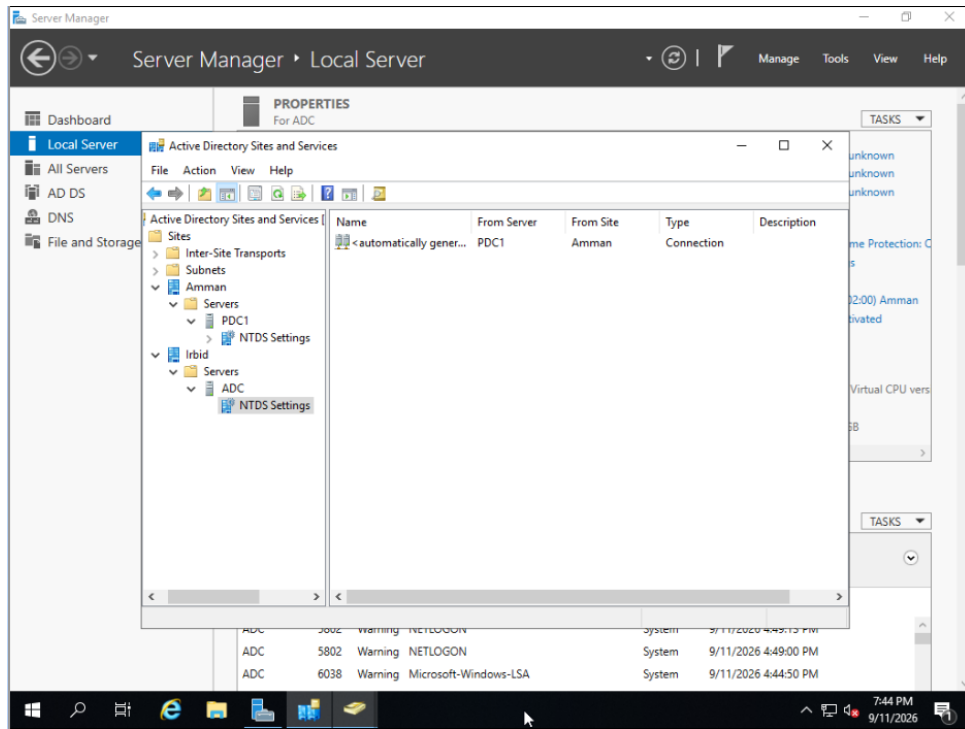


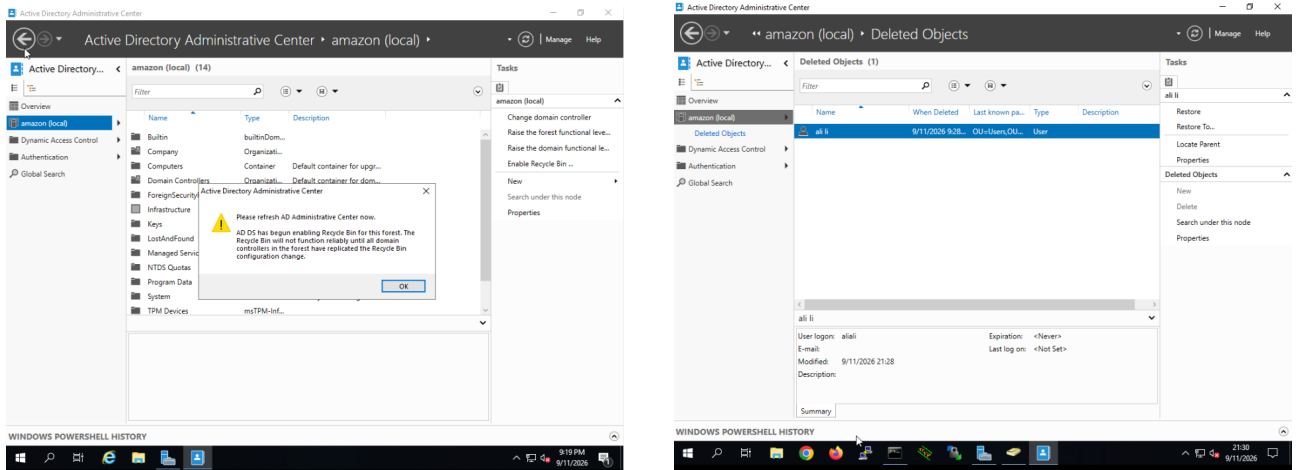
Figure 7: Geographic site partitioning in Active Directory Sites and Services showing the Amman and Irbid hierarchy.

15.1 Active Directory Recycle Bin Architecture and Activation

To prevent destructive directory loss from accidental deletion, the **Active Directory Recycle Bin** optional feature was committed at the forest scope. In standard directory architectures, deleted objects instantly strip structural attributes—including group nesting (`memberOf`) and security identifiers (`objectSid`)—entering an irreversible *Tombstone* state that necessitates offline Directory Services Restore Mode (DSRM) authoritative restores. Activating the Recycle Bin alters the underlying `ntds.dit` object lifecycle: objects marked with `isDeleted=TRUE` preserve 100% of their DACLs, SID baselines, and attributes inside the logical `CN=Deleted Objects` container across the `msDS-deletedObjectLifetime` retention window (180 days).

15.1.1 Provisioning and Lifecycle Verification

- **Feature Enablement:** The forest functional level was validated at Windows Server 2016 baseline[cite: 1]. The feature was executed via Active Directory Administrative Center (`dsac.exe`) against the root domain boundary, permanently binding the container hierarchy.
- **Disaster Simulation & Recovery:** A verification principal (`JuniorAdmin`) assigned to `IT-Staff` within `OU=IT,OU=Company` was created and deleted. The tombstone bypass was validated by retrieving the tombstoned object from `Deleted Objects` and initiating an in-place `Restore`.
- **Integrity Audit:** Post-recovery inspection confirmed zero attribute degradation: the restored identity retained its original `objectSid` and instant membership in `IT-Staff` without administrative intervention.



(a) Recycle Bin feature enablement confirmation in ADAC.

(b) Deleted Objects container and restoration audit.

Figure 8: Active Directory Recycle Bin activation and object recovery verification matrix.

15.2 High Availability DHCP Deployment and Cross-VLAN Failover

To eliminate single-point-of-failure vulnerabilities in dynamic host configuration across the client infrastructure, a high-availability **DHCP Failover** architecture was deployed in **Hot Standby** mode between the primary domain controller (DC01: 192.168.10.10) and the secondary server (ADC: 192.168.10.11).

15.2.1 Protocol Architecture and State Machine Mechanics

- **Synchronization Protocol (TCP 647):** The failover partners establish an authenticated, persistent TCP control session over port 647 to synchronize client lease bindings (BINDING-UPDATE / BINDING-ACK) and maintain mutual heartbeat monitoring.
- **Operational State Model:** In standard state (**Normal**), the active node (DC01) services 100% of client leases across Scope 192.168.20.0/24. The secondary server (ADC) maintains a standby posture with a dedicated **5% Reserve Pool**.
- **Automated Failover Transition:** Upon failure of DC01, the standby node detects dropped heartbeat packets and transitions to **Communications Interrupted**, servicing renewals and discoveries from its reserve partition. Following the expiration of the **State Switchover Interval** (10 minutes), ADC transitions automatically to **Partner Down**, assuming full 100% leasing authority over the entire subnet address space without administrative intervention.
- **Dual IP Helper Relay Forwarding:** To enable cross-VLAN lease negotiation under failover conditions, the Layer 3 Core Switch SVI (Vlan20) was configured with dual forwarding entries. Upon receiving a client broadcast DHCPDISCOVER (UDP 67), the switch intercepts the frame, injects its SVI address (192.168.20.1) into the giaddr field, and replicates the payload as two discrete unicast transmissions targeted to 192.168.10.10 and 192.168.10.11 simultaneously.

15.2.2 Device and Service Configuration

The secondary server was provisioned with the DHCP Server role and authorized within Active Directory (CN=NetServices):

```
1 Install-WindowsFeature -Name DHCP -IncludeManagementTools
2 Add-DhcpServerInDC -DnsName "ADC.amazon.local" -IPAddress 192.168.10.11
```

Listing 1: DHCP Role Authorization on Secondary Domain Controller

On the primary server (DC01), the failover relationship was instantiated across the target client scope with the parameters defined in Table 8:

Parameter	Configured Value	Operational Function
Relationship Name	DC01-ADC-VLAN20	Failover association identifier
Operational Mode	Hot Standby	Active/Passive operational split
Partner Server Role	Standby	Designates ADC as passive recovery node
Reserve Percentage	5%	Unassigned pool for emergency lease issuance
Maximum Client Lead Time	01:00:00 (1 Hour)	Lease extension window during state drift
State Switchover Interval	10 Minutes	Automated transition timer to Partner Down
Authentication	Enabled (Shared Secret)	Cryptographic packet verification (TCP 647)

Table 8: DHCP Failover operational parameters and timing baselines.

The Layer 3 switch relay configuration was expanded to maintain redundant forwarders:

```

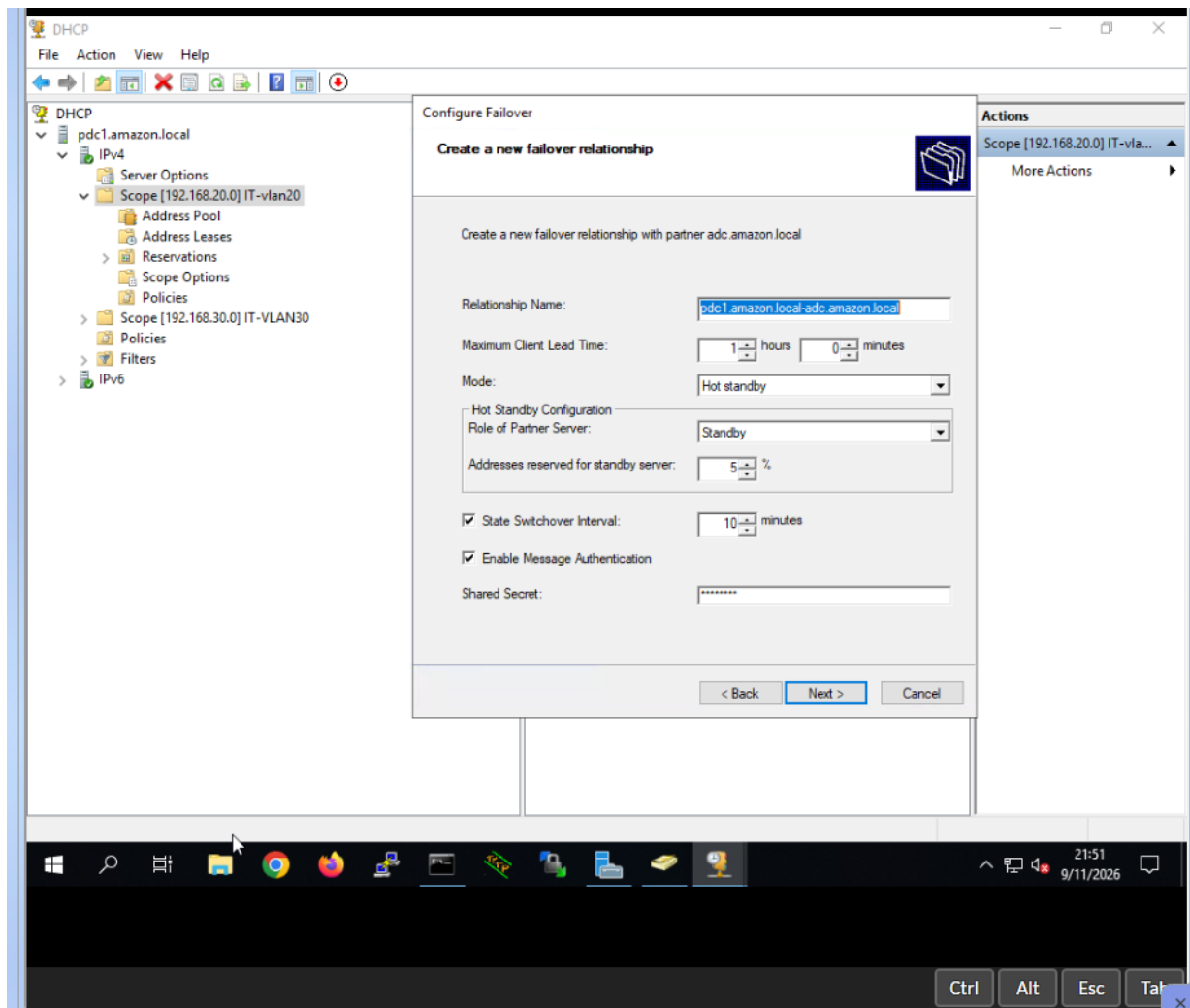
1 interface Vlan20
2   ip address 192.168.20.1 255.255.255.0
3   ip helper-address 192.168.10.10
4   ip helper-address 192.168.10.11

```

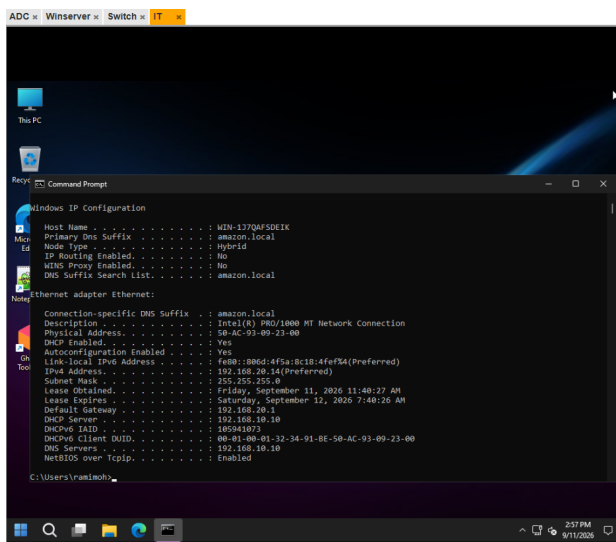
Listing 2: L3 Core Switch Dual Helper-Address Configuration

15.2.3 Implementation Evidence and Outage Simulation

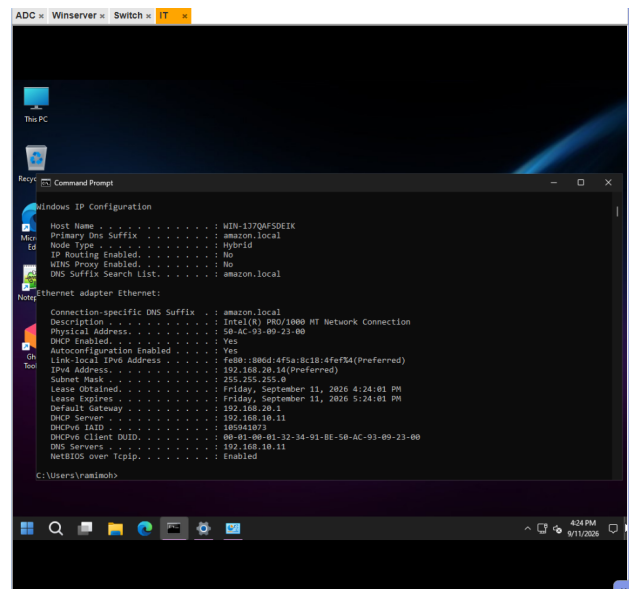
Operational continuity was validated through physical fault injection by isolating DC01 from the network. Client workstations executed forced lease release and renewal cycles (`ipconfig /release` followed by `ipconfig /renew`). Inspection of network adapter properties verified that dynamic address provisioning transferred from the primary to the standby controller without interruption.



(a) DHCP Failover relationship configuration parameters, Hot Standby mode, and 5% reserve allocation.



(b) Baseline lease validation showing address assignment from DC01 (192.168.10.10).



(c) Post-outage lease renewal confirming active failover response from ADC (192.168.10.11).

Figure 9: End-to-end verification of DHCP Failover state transition, dual relay routing, and automated service recovery.

16 Active Directory Disaster Recovery: FSMO Role Seizure and Core Services Remediation

16.1 Architectural Overview and Failure Modeling

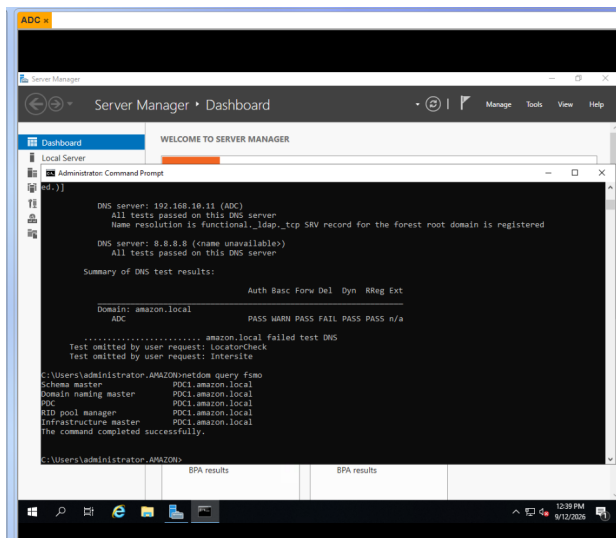
Active Directory Domain Services (AD DS) relies on a multi-master replication model for most directory updates, while operations requiring strict conflict avoidance are governed by five Flexible Single Master Operation (FSMO) roles. In this enterprise infrastructure, all five authorities were originally centralized on the primary domain controller (PDC1.amazon.local, 192.168.10.10):

- **Forest-Wide Roles:** *Schema Master* (structural updates to object classes and attributes) and *Domain Naming Master* (forest domain namespace additions and removals).
- **Domain-Wide Roles:** *PDC Emulator* (Kerberos password updates, time synchronization, and account lockouts), *RID Master* (Relative Identifier pool allocation to prevent SID collisions), and *Infrastructure Master* (cross-domain object reference translation).

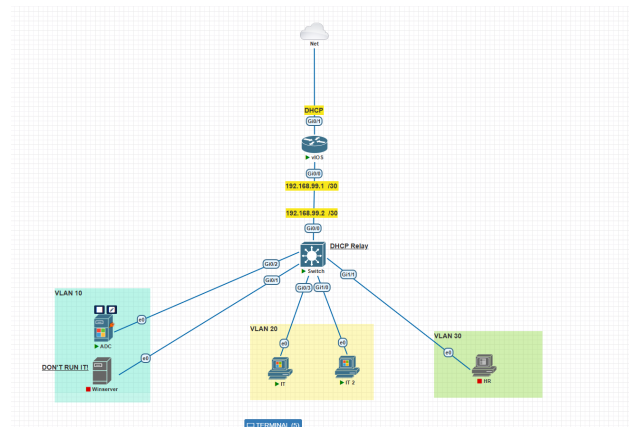
A catastrophic hardware failure was simulated by severing the hypervisor power state to PDC1, placing it in a permanently unrecoverable state. Under routine operational conditions, role reassignment occurs via mutual Remote Procedure Call (RPC) synchronization (Transfer). Because the RPC endpoint was unreachable, graceful transfer failed. The secondary domain controller (ADC.amazon.local, 192.168.10.11) executed a unilateral database claim (Seizure) to assume operations master authority and restore directory write capabilities.

16.2 Pre-Outage Baseline and Fault Injection

Prior to fault injection, directory replication status and FSMO role allocations were validated using the `netdom query fsmo` command. All five roles pointed authoritatively to PDC1.amazon.local. Following baseline documentation, PDC1 was powered down.



(a) Baseline query confirming PDC1 role ownership.



(b) Permanent offline state of PDC1 in virtualization lab.

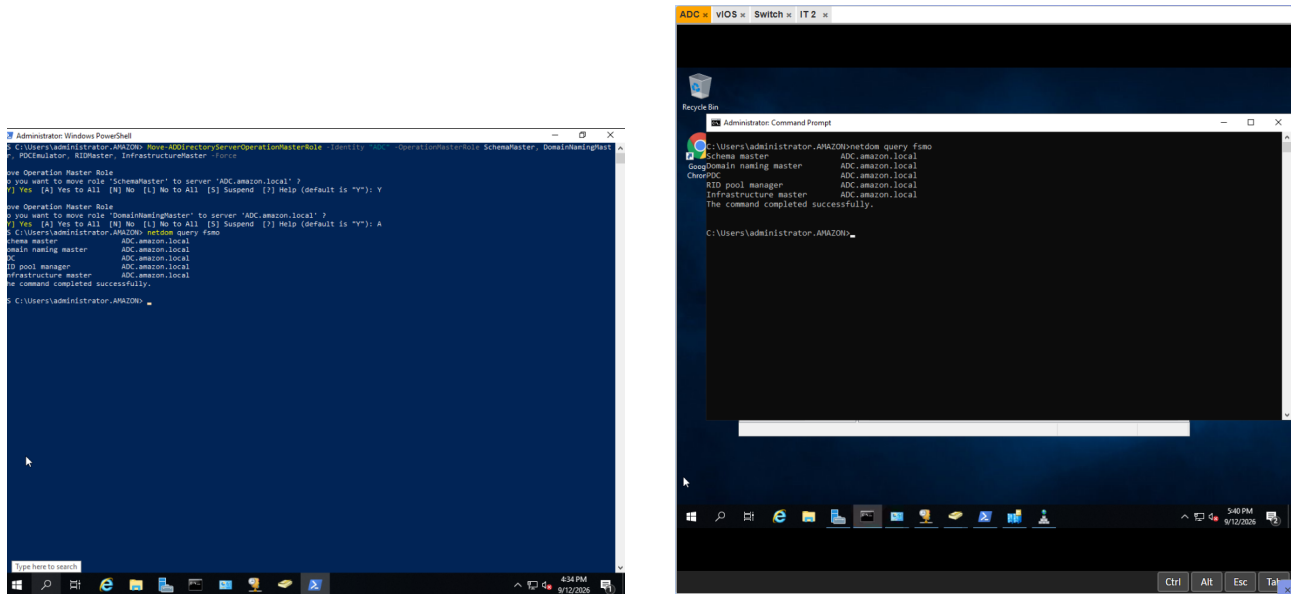
Figure 10: Pre-failure operational baseline and disaster simulation.

16.3 Forced Operation Master Role Seizure

Graphical administration tools (ADUC, Active Directory Domains and Trusts, and the registered Active Directory Schema snap-in via `regsvr32 schmmgmt.dll`) stalled on RPC timeout limits when the target host failed to respond. To prevent indefinite management deadlocks, the seizure routine was executed directly through PowerShell on ADC:

```
1 Move-ADDirectoryServerOperationMasterRole -Identity "ADC" -OperationMasterRole SchemaMaster, DomainNamingMaster, PDCEmulator, RIDMaster, InfrastructureMaster -Force
```

The `-Force` parameter overrode the requirement for an online RPC handshake with the previous role holder. The secondary server modified its local `ntds.dit` database partitions directly, writing its own NTDS Settings GUID to the `fsmoRoleOwner` attribute across the Schema, Configuration, and Domain naming contexts. The RID Master seizure advanced the Relative Identifier pool by a designated block to guarantee that future security principal creation would avoid SID collisions.



(a) Execution of role seizure using PowerShell with `-Force` override.

(b) Post-seizure validation displaying `ADC.amazon.local` as authoritative owner.

Figure 11: Operation Master seizure execution and database ownership verification.

16.4 Directory Metadata and Stale DNS Record Purging

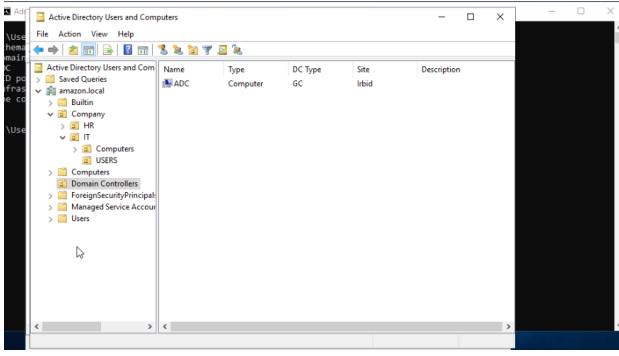
Seizing FSMO roles does not purge the failed domain controller object from the Active Directory topology. Retaining orphaned server references forces the Knowledge Consistency Checker to log periodic replication failures and misdirects domain locator traffic. Metadata cleanup was completed using the following operations:

1. **Directory Object Removal:** Inside *Active Directory Users and Computers*, the PDC1 computer object was deleted from `OU=Domain Controllers`. Checking the prompt specifying that the domain controller was permanently offline triggered the automated metadata cleanup mechanism, stripping the server's directory references.
2. **Replication Link Purging:** In *Active Directory Sites and Services*, the PDC1 server object was deleted from `Sites → Amman → Servers` to terminate invalid replication connection links.
3. **DNS Zone Remediation:** In *DNS Manager*, obsolete `Host (A)` records and `Name Server (NS)` entries pointing to `192.168.10.10` were deleted from `amazon.local` and `_msdcs.amazon.local`. The stale storage alias `files.amazon.local (CNAME)` pointing to PDC1 was also removed to prevent network blackholing.

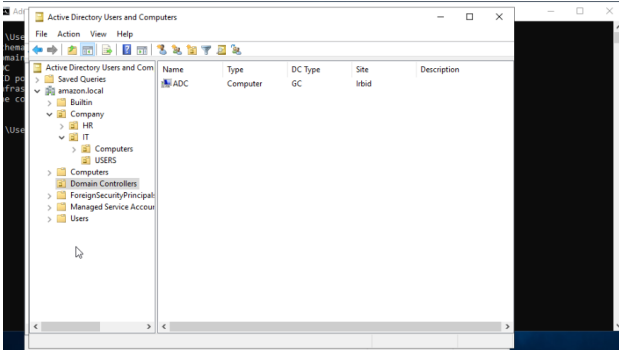
16.5 Core Services Remediation and Workstation Verification

Because PDC1 also functioned as the primary DNS resolver, workstations relying on DHCP lease options were unable to perform name resolution following the outage. Service continuity was restored via the following implementations:

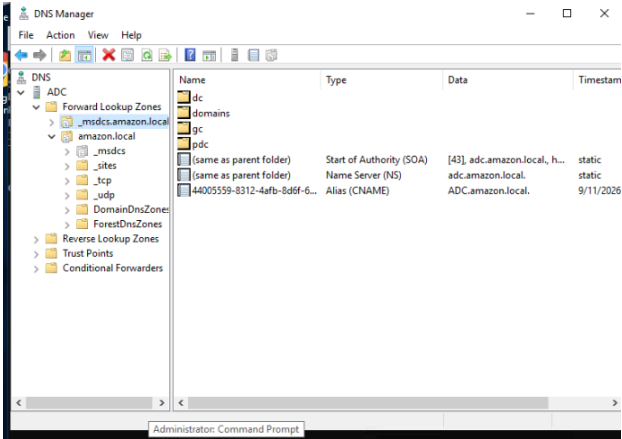
- **DHCP Scope Option 006 Reconfiguration:** Scope Option 006 (DNS Servers) on the active DHCP service hosted on ADC was updated to distribute `192.168.10.11` as the primary resolver, removing the decommissioned `192.168.10.10` address.
- **DHCP Failover State Mechanics:** Following TCP 647 sync session termination, ADC entered the *Communications Interrupted* state. The service serviced DHCP requests from its 5% reserve allocation using a 1-hour Maximum Client Lead Time (MCLT) lease window.



(a) Purging failed DC object from Domain Controllers container.



(b) Purging orphaned NTDS connection objects from Amazon site.



(c) Purging stale Name Server (NS) delegation records.

Figure 12: Active Directory metadata cleanup and DNS zone remediation matrix (2 × 2).

- **Workstation Lease Renewal:** Workstation PC-IT2 executed `ipconfig /release` followed by `ipconfig /renew`.

As shown in Figure 13, the client obtained IP 192.168.20.146 from the ADC DHCP server (192.168.10.11) across the Layer 3 switch relay. The lease duration matched the configured 1-hour MCLT window (4 : 54 : 55 PM → 5 : 54 : 57 PM), validating operational failover state enforcement. Domain authentication succeeded under the user context `amazon.local\ramimoh`.

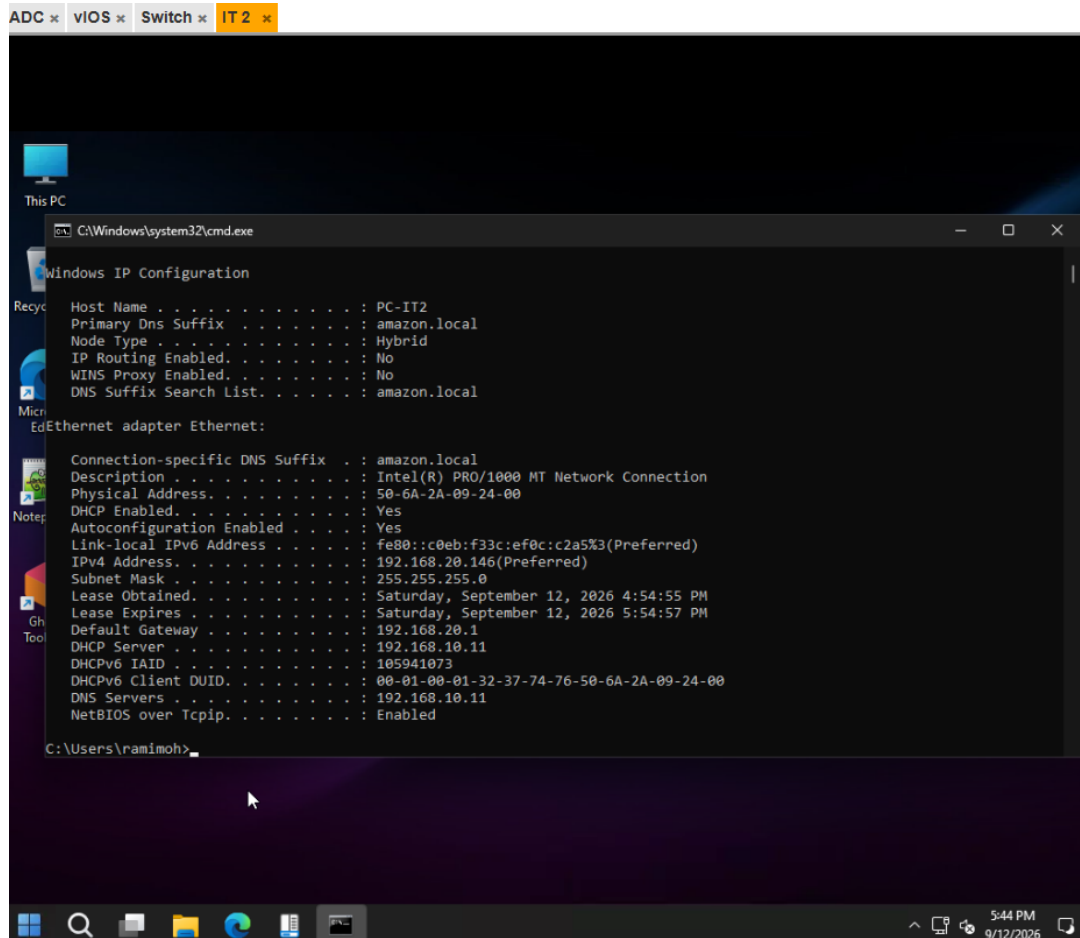


Figure 13: Workstation PC-IT2 verification: dynamic lease allocation from 192.168.10.11, updated DNS assignment, 1-hour MCLT enforcement, and successful user login.

16.6 Directory Write Verification and System Diagnostics

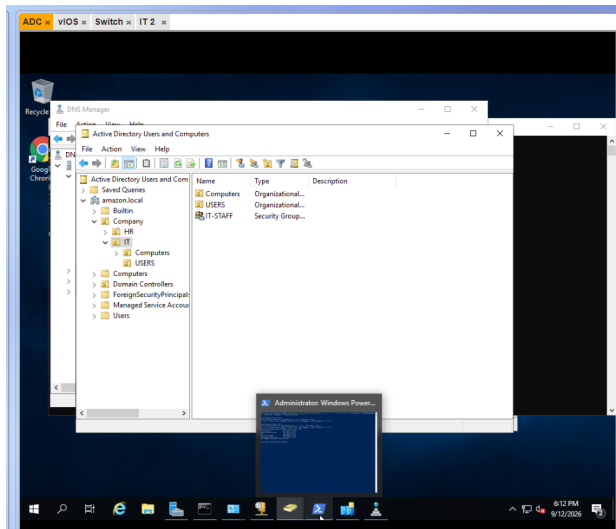
To confirm that the RID Master seizure allowed ADC to autonomously issue Relative Identifiers, a test user (`TestUser`) and global security group (`TestGroup`) were provisioned in the directory. Both objects committed immediately, confirming RID pool initialization and schema operational readiness.

Final directory and name resolution health checks were conducted using administrative utilities:

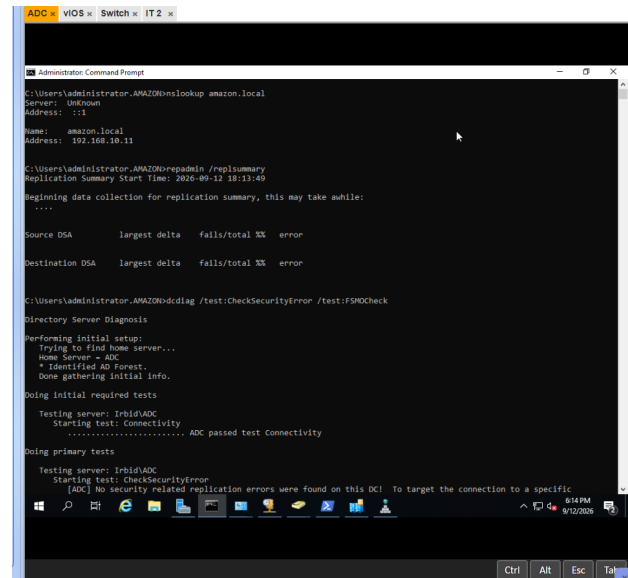
- `nslookup amazon.local`: Resolved to 192.168.10.11 without timeout or reference to the failed server.
- `repadmin /replsummary`: Confirmed zero synchronization failures or lingering outbound replication attempts.
- `dcdiag /v`: Validated critical operational tests including Advertising, FSMOCheck, and MachineAccount.

16.7 Diagnostic Log: Faults Encountered and Engineering Resolutions

Table 9 outlines the operational issues encountered during deployment, their underlying technical causes, and the corrective engineering procedures applied.



(a) Provisioning **TestUser** and **TestGroup** verifying active RID pool.



(b) Consolidated DNS resolution and clean diagnostic check output.

Figure 14: Directory write operational testing and diagnostic verification.

Observed Issue	Underlying Mechanism / Cause	Corrective Action
SVI Vlan10 operational state dropped to down/down.	Cisco <i>Autostate</i> algorithm suppressed the SVI because no physical access ports were actively forwarding inside VLAN 10 after node startup without persisted running-config.	Re-assigned interfaces g0/1 and g0/2 to VLAN 10 in access mode and enabled immediate Spanning Tree forwarding via spanning-tree portfast .
GUI Operation Master transfer hung indefinitely.	Management snap-ins require active synchronous RPC communication with the current FSMO holder before presenting forced transfer dialogs.	Bypassed RPC timeouts by executing Move-ADDirectoryServerOperationMasterRole via PowerShell using the -Force parameter.
Workstations failed domain DNS resolution post-outage.	The active DHCP scope distributed the decommissioned domain controller (192.168.10.10) under Scope Option 006.	Reconfigured Option 006 on ADC to 192.168.10.11 and forced client lease renegotiation via ipconfig /renew .

Table 9: Operational troubleshooting log and engineering remediation steps.

16.8 Assessment Verification

The disaster recovery deployment satisfies all verification criteria defined in the laboratory engineering specification, as documented in Table 10.

Assessment Item	Status	Technical Evidence Reference
FSMO Role Ownership	PASSED	ADC.amazon.local confirmed as owner across all five roles via <code>netdom query fsmo</code> .
AD DS Operational Health	PASSED	Successful object creation (<code>TestUser/TestGroup</code>) and clean <code>dcdiag /v</code> output.
DNS Architecture Integrity	PASSED	Stale PDC1 records removed; domain apex resolves cleanly to 192.168.10.11.
Client Authentication	PASSED	Workstation PC-IT2 validated lease from ADC and authenticated user <code>ramimoh</code> .
Metadata Cleanup	PASSED	Server object purged from ADUC, Sites and Services, and DNS zone records.

Table 10: Disaster recovery assessment checklist and compliance evidence.